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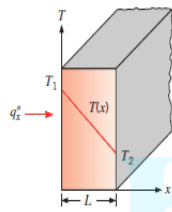
Conduction

Conduction is the transfer of energy from the more energetic to the less energetic particles of a substance due to interactions between the particles. Conduction can take place in solids, liquids, or gases. In gases and liquids, conduction is due to the collisions and diffusion of the molecules during their random motion. In solids, it is due to the combination of vibrations of the molecules in a lattice and the energy transport by free electrons.

Define Fourier law of heat conduction. What are the key features of Fourier law?

Fourier's law of heat conduction states that the rate of heat transfer through a material is proportional to the negative gradient in the temperature and the area (perpendicular to the gradient) of the surface through which the heat flows.

For the one-dimensional plane wall shown in Figure on the right, having a temperature distribution $T(x)$, the rate equation is expressed as,



$$q''_x = -k \frac{dT}{dx} A$$

q''_x - heat flux or heat transfer rate per unit area normal to the direction of heat transfer

The parameter k is a transport property known as the thermal conductivity (W/m K) and is a characteristic of the wall material. The minus sign indicates that heat is transferred in the direction of decreasing temperature.

Recognizing that the heat flux is a vector quantity, we can write a more general statement of the conduction rate equation (Fourier's law) as follows:

$$\mathbf{q}'' = -k \nabla T A = -k \left(i \frac{\partial T}{\partial x} + j \frac{\partial T}{\partial y} + k \frac{\partial T}{\partial z} \right) A$$

Key features of Fourier's law

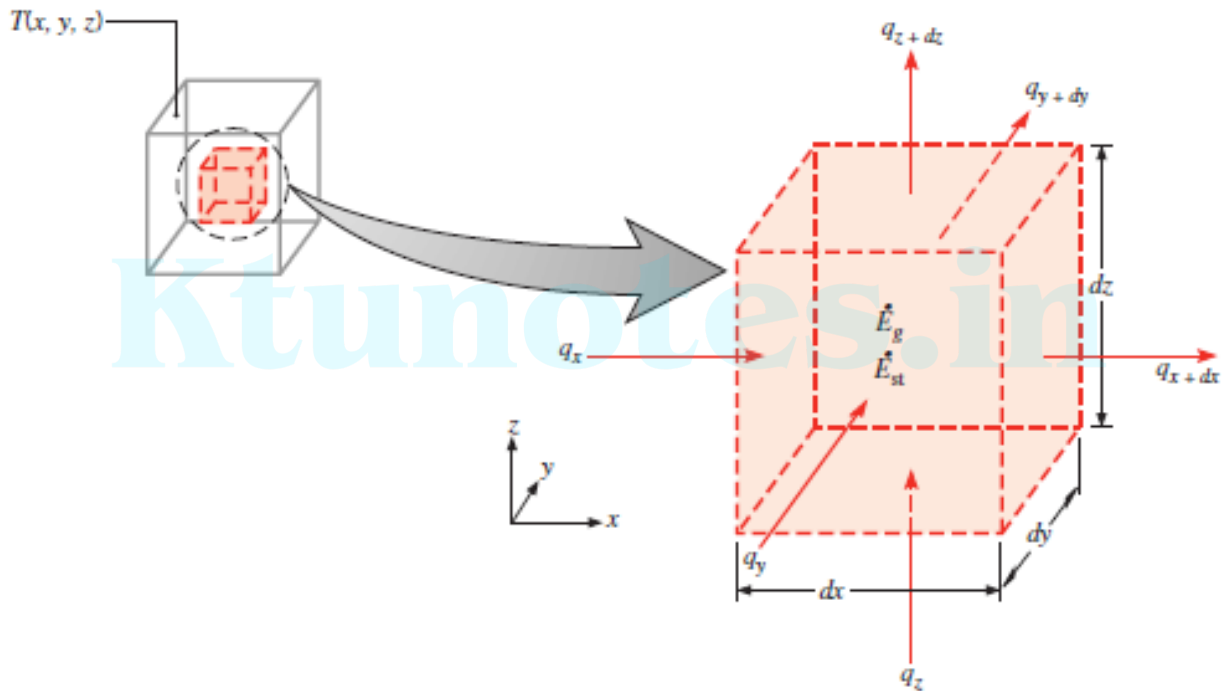
- It is not an expression that may be derived from first principles; it is instead a generalization based on experimental evidence.
- It defines an important material property, the thermal conductivity.
- The heat flux is normal to an isotherm and in the direction of decreasing temperature.
- The Fourier's law applies for all matter, regardless of its state (solid, liquid, or gas).

The Heat Diffusion Equation (Cartesian coordinates)

Consider a homogeneous medium within which there is no bulk motion (advection) and the temperature distribution $T(x, y, z)$ is expressed in cartesian coordinates. We define an infinitesimally small (differential) control volume, $dx.dy.dz$, as shown in figure below.

Apply the rate form of the first law to the control volume. Only thermal forms of energy will be considered. In the absence of motion (or with uniform motion), there are no changes in mechanical energy and no work being done on the system. If there are temperature gradients, conduction heat transfer will occur across each of the control surfaces.

The conduction heat rates perpendicular to each of the control surfaces at the x -, y -, and z -coordinate locations are indicated by the terms q_x , q_y , and q_z , respectively



The conduction heat rates at the opposite surfaces can then be expressed as a Taylor series expansion where, neglecting higher-order terms,

$$q_{x+dx} = q_x + \frac{\partial q_x}{\partial x} dx \quad q_{y+dy} = q_y + \frac{\partial q_y}{\partial y} dy \quad q_{z+dz} = q_z + \frac{\partial q_z}{\partial z} dz \quad (1)$$

[The above equation for q_{x+dx} , state that the x - component of the heat transfer rate at $x+dx$ is equal to the value of this component at x plus the amount by which it changes with respect to x (i.e. $\frac{\partial q_x}{\partial x}$) times dx .]

Within the medium there may also be an energy source term associated with the rate of thermal energy generation. This term is represented as,

$$\dot{E}_g = \dot{q} \, dx \, dy \, dz \quad (2)$$

[The energy generation term represents some energy conversion process involving thermal energy on one hand and some other form of energy, such as chemical, electrical, or nuclear, on the other.]

where, $\dot{q}$ is the rate at which energy is generated per unit volume of the medium (W/m^3). If the material is not experiencing a change in phase, latent energy effects shall be neglected, and the energy storage term may be expressed as,

$$\dot{E}_{st} = \rho c_p \frac{\partial T}{\partial t} \, dx \, dy \, dz \quad (3)$$

On a rate basis, the general form of the conservation of energy requirement is

$$\dot{E}_{in} - \dot{E}_{out} + \dot{E}_g = \dot{E}_{st} \quad (4)$$

$$\underbrace{(q_x + q_y + q_z)}_{\dot{E}_{in}} - \underbrace{(q_{x+dx} + q_{y+dy} + q_{z+dz})}_{\dot{E}_{out}} + \underbrace{\dot{q} \, dx \, dy \, dz}_{\dot{E}_g} = \underbrace{\rho c_p \frac{\partial T}{\partial t} \, dx \, dy \, dz}_{\dot{E}_{st}} \quad (5)$$

$$-\left(\underbrace{\frac{\partial q_x}{\partial x} dx + \frac{\partial q_y}{\partial y} dy + \frac{\partial q_z}{\partial z} dz}_{\dot{E}_{in} - \dot{E}_{out}} \right) + \underbrace{\dot{q} \, dx \, dy \, dz}_{\dot{E}_g} = \underbrace{\rho c_p \frac{\partial T}{\partial t} \, dx \, dy \, dz}_{\dot{E}_{st}} \quad (5)$$

The heat transfer in x, y and z directions (q_x , q_y , and q_z respectively) can be expressed in terms of the corresponding temperature gradients by invoking Fourier's law of heat conduction (assuming an isotropic medium).

$$q_x = -k \, dy \, dz \frac{\partial T}{\partial x} \quad q_y = -k \, dz \, dx \frac{\partial T}{\partial y} \quad q_z = -k \, dx \, dy \frac{\partial T}{\partial z} \quad (6)$$

Substituting for q_x , q_y and q_z in eq. 5,

$$\frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t} \quad (7)$$

Equation 7 is the general form, in Cartesian coordinates, of the heat diffusion equation or heat equation. From its solution, we can obtain the temperature distribution $T(x, y, z)$ as a function of time. The heat equation, Equation 7, states that *at any point in the medium the net rate of energy transfer by conduction into a unit volume plus the volumetric rate of thermal energy generation must equal the rate of change of thermal energy stored within the volume.*

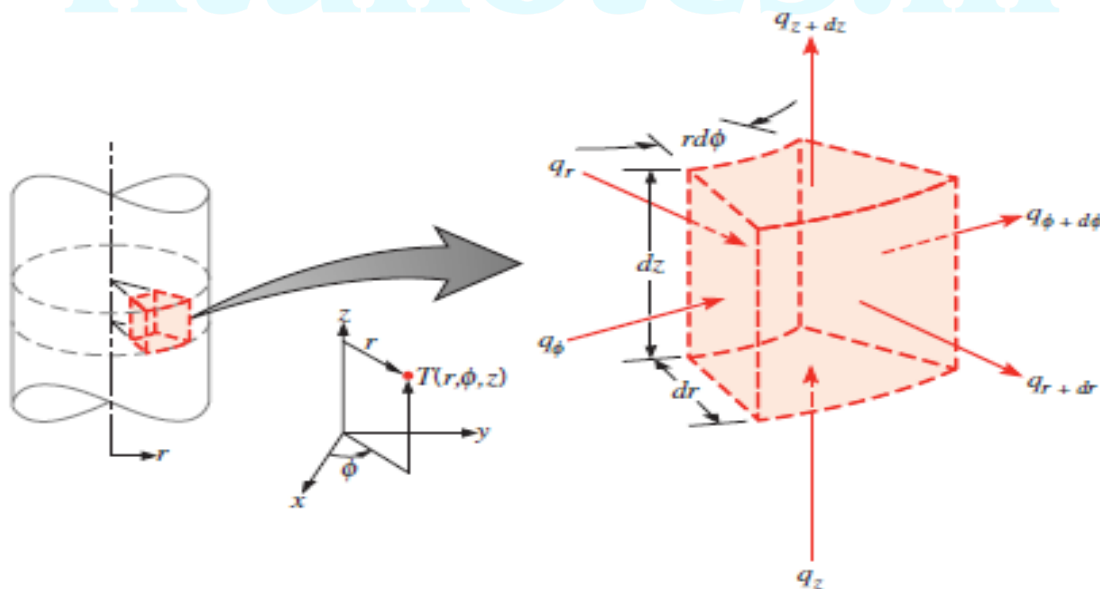
If the thermal conductivity is constant, the heat equation can be simplified to,

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad (8)$$

where $\alpha = \frac{k}{\rho c_p}$ is the thermal diffusivity (m^2/s). It measures the ability of a material to conduct thermal energy relative to its ability to store thermal energy. Materials of large α will respond quickly to changes in their thermal environment, whereas materials of small α will respond more sluggishly, taking longer to reach a new equilibrium condition.

The heat equation in cylindrical coordinates

Consider a homogeneous medium within which there is no bulk motion (advection) and the temperature distribution $T(r, \phi, z)$ is expressed in cylindrical coordinates. We define an infinitesimally small (differential) control volume, $dr \cdot d\phi \cdot dz$, as shown in figure below.



Apply the rate form of the first law to the control volume. Only thermal forms of energy will be considered. In the absence of motion (or with uniform motion), there are no changes in mechanical energy and no work being done on the system. If there are temperature gradients, conduction heat transfer will occur across each of the control surfaces.

The conduction heat rates perpendicular to each of the control surfaces at the r -, ϕ -, and z -coordinate locations are indicated by the terms q_r , q_ϕ , and q_z , respectively

The conduction heat rates at the opposite surfaces can then be expressed as a Taylor series expansion where, neglecting higher-order terms,

$$q_{r+dr} = q_r + \frac{\partial q_r}{\partial r} dr \quad q_{\phi+d\phi} = q_\phi + \frac{1}{r} \frac{\partial q_\phi}{\partial \phi} r d\phi \quad q_{z+dz} = q_z + \frac{\partial q_z}{\partial z} dz$$

Within the medium there may also be an energy source term associated with the rate of thermal energy generation. This term is represented as,

$$\dot{E}_g = \dot{q} r dr d\phi dz \quad (2)$$

[The energy generation term represents some energy conversion process involving thermal energy on one hand and some other form of energy, such as chemical, electrical, or nuclear, on the other.]

where, $\dot{q}$ is the rate at which energy is generated per unit volume of the medium (W/m^3). If the material is not experiencing a change in phase, latent energy effects shall be neglected, and the energy storage term may be expressed as,

$$\dot{E}_{st} = \rho c_p \frac{\partial T}{\partial t} r dr d\phi dz$$

On a rate basis, the general form of the conservation of energy requirement is

$$\dot{E}_{in} - \dot{E}_{out} + \dot{E}_g = \dot{E}_{st} \quad (4)$$

$$\underbrace{(q_r + q_\phi + q_z)}_{\dot{E}_{in}} - \underbrace{(q_{r+dr} + q_{\phi+d\phi} + q_{z+dz})}_{\dot{E}_{out}} + \underbrace{\dot{q} r dr d\phi dz}_{\dot{E}_g} = \underbrace{\rho c_p \frac{\partial T}{\partial t} r dr d\phi dz}_{\dot{E}_{st}}$$

$$-\left(\underbrace{\frac{\partial q_r}{\partial r} dr + \frac{\partial q_\phi}{\partial \phi} d\phi + \frac{\partial q_z}{\partial z} dz}_{\dot{E}_{in} - \dot{E}_{out}} \right) + \underbrace{\dot{q} r dr d\phi dz}_{\dot{E}_g} = \underbrace{\rho c_p \frac{\partial T}{\partial t} r dr d\phi dz}_{\dot{E}_{st}} \quad (5)$$

The heat transfer in x , y and z directions (q_x , q_y , and q_z respectively) can be expressed in terms of the corresponding temperature gradients by invoking Fourier's law of heat conduction (assuming an isotropic medium).

$$q_x = -k r d\phi dz \frac{\partial T}{\partial r} \quad q_y = -k dr dz \frac{1}{r} \frac{\partial T}{\partial \phi} \quad q_z = -k r d\phi dr \frac{\partial T}{\partial z}$$

Substituting for q_r , q_ϕ and q_z in eq. 5,

$$\frac{1}{r} \frac{\partial}{\partial r} \left(kr \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \phi} \left(k \frac{\partial T}{\partial \phi} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t} \quad (7)$$

Equation 7 is the general form, in cylindrical coordinates, of the heat diffusion equation or heat equation. From its solution, we can obtain the temperature distribution $T(r, \phi, z)$ as a function of time. The heat equation, Equation 7, states that *at any point in the medium the net rate of energy transfer by conduction into a unit volume plus the volumetric rate of thermal energy generation must equal the rate of change of thermal energy stored within the volume.*

If the thermal conductivity is constant, the heat equation can be simplified to,

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \phi^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad (8)$$

where $\alpha = \frac{k}{\rho c_p}$ is the thermal diffusivity (m^2/s). It measures the ability of a material to conduct thermal energy relative to its ability to store thermal energy. Materials of large α will respond quickly to changes in their thermal environment, whereas materials of small α will respond more sluggishly, taking longer to reach a new equilibrium condition.

One dimensional heat conduction without heat generation

Plane wall

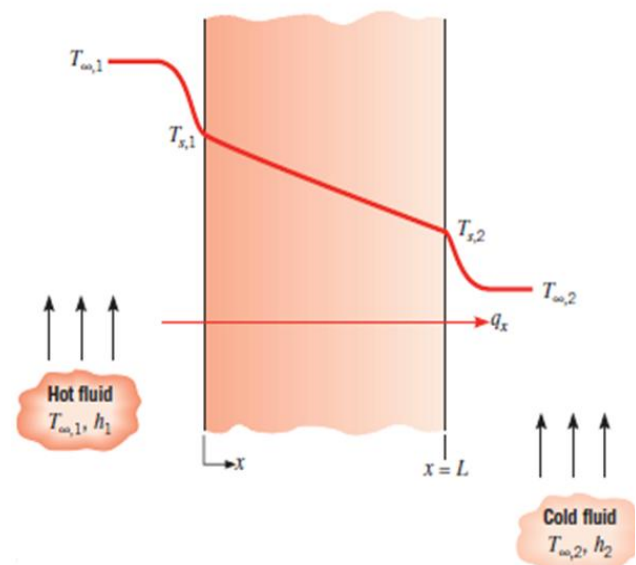
The temperature distribution in the wall can be determined by solving the heat equation with the proper boundary conditions. For steady-state conditions with no heat generation or heat absorption within the wall, the appropriate form of the heat equation is

$$\frac{d}{dx} \left(k \frac{dT}{dx} \right) = 0$$

Applying the specified temperature boundary condition (Dirichlet boundary condition) at $x=0$ and $x=L$.

$$T(x=0) = T_{s,1}$$

$$T(x=L) = T_{s,2}$$



The temperature distribution in the wall is

$$T(x) = (T_{s,2} - T_{s,1}) \frac{x}{L} + T_{s,1}$$

Once the temperature distribution in the wall is known, the heat transfer through the wall can be obtained by using Fourier's law.

$$q_x = -kA \frac{dT}{dx} = \frac{kA}{L} (T_{s,1} - T_{s,2})$$

Defining resistance as the ratio of a driving potential to the corresponding transfer rate, it follows from the above equation that the thermal resistance for conduction in a plane wall is,

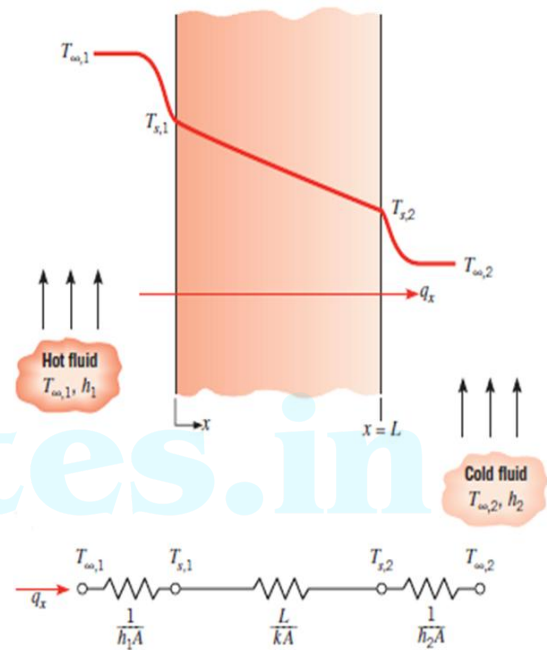
$$R_{t,cond} = \frac{T_{s,1} - T_{s,2}}{q_x} = \frac{L}{kA}$$

A thermal resistance may also be associated with heat transfer by convection at a surface. The convection heat transfer from a surface to a fluid is

$$q_{conv} = hA(T_s - T_\infty)$$

The thermal resistance for convection is then,

$$R_{t,conv} = \frac{T_s - T_\infty}{q_x} = \frac{1}{hA}$$



Cylindrical wall

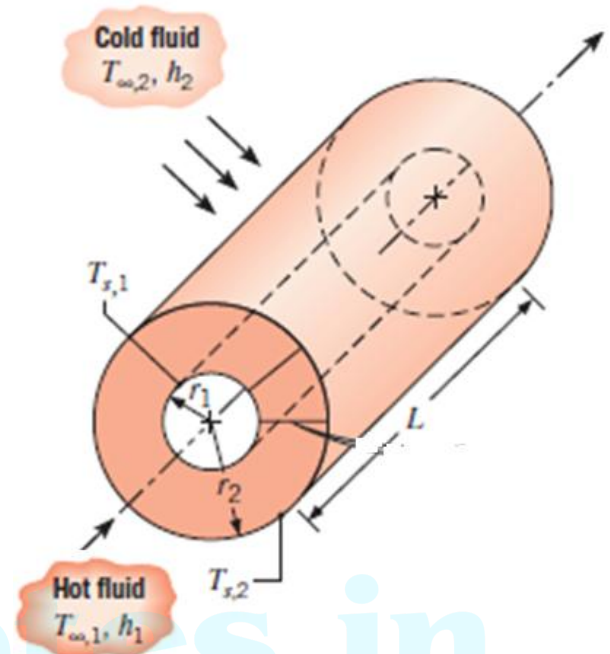
The temperature distribution in the wall can be determined by solving the heat equation with the proper boundary conditions. For steady-state conditions with no heat generation or heat absorption within the wall, the appropriate form of the heat equation is

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dT}{dr} \right) = 0$$

Applying the specified temperature boundary condition (Dirichlet boundary condition) at $r=r_1$ and $r=r_2$.

$$T(r = r_1) = T_{s,1}$$

$$T(r = r_2) = T_{s,2}$$



The temperature distribution in the wall is

$$T(r) = \frac{T_{s,1} - T_{s,2}}{\ln\left(\frac{r_1}{r_2}\right)} \ln\left(\frac{r}{r_2}\right) + T_{s,2}$$

Once the temperature distribution in the wall is known, the heat transfer through the wall can be obtained by using Fourier's law.

$$q_x = -k(2\pi r_1 L) \frac{dT}{dr} \Big|_{r=r_1} = \frac{2\pi k L (T_{s,1} - T_{s,2})}{\ln\left(\frac{r_2}{r_1}\right)}$$

From this result it is evident that, for radial conduction in a cylindrical wall, the thermal resistance is of the form

$$R_{t,cond} = \frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi k L}$$

Spherical wall

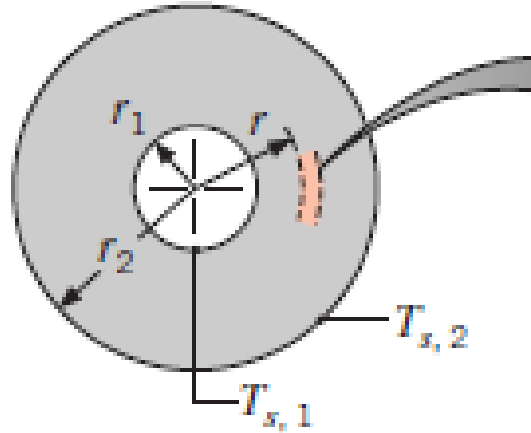
The temperature distribution in the wall can be determined by solving the heat equation with the proper boundary conditions. For steady-state conditions with no heat generation or heat absorption within the wall, the appropriate form of the heat equation is

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = 0$$

Applying the specified temperature boundary condition (Dirichlet boundary condition) at $r=r_1$ and $r=r_2$.

$$T(r = r_1) = T_{s,1}$$

$$T(r = r_2) = T_{s,2}$$



The temperature distribution in the wall is

$$T(r) = \frac{T_{s,1} - T_{s,2}}{(r_2 - r_1)} \frac{1}{r} + \frac{r_2 T_{s,2} - r_1 T_{s,1}}{(r_2 - r_1) r_1 r_2}$$

Once the temperature distribution in the wall is known, the heat transfer through the wall can be obtained by using Fourier's law.

$$q_x = -k(4\pi r_1^2) \frac{dT}{dr} \Big|_{r=r_1} = (4\pi r_1 r_2 k) \frac{T_{s,1} - T_{s,2}}{(r_2 - r_1)}$$

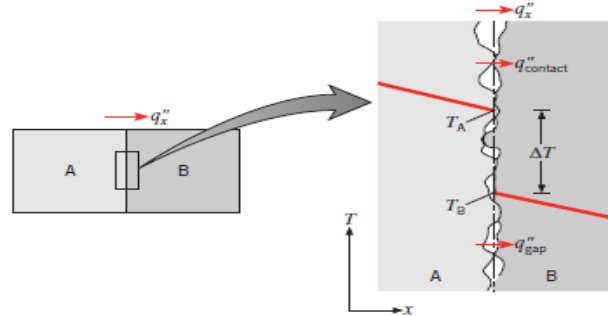
From this result it is evident that, for radial conduction in a spherical wall, the thermal resistance is of the form

$$R_{t,cond} = \frac{(r_2 - r_1)}{4\pi r_1 r_2 k}$$

What is thermal contact resistance? How can thermal contact resistance be minimized?

When two surfaces are brought in contact, the contact points are interspersed with gaps (due to surface roughness effects) that are, in most instances, air filled. Heat transfer is therefore due to conduction across the actual contact area and to conduction and/or radiation across the gaps. This results in a thermal resistance at the contact interface, leading to a temperature drop across the interface. This resistance is known as the thermal contact resistance, $R_{t,c}$. For a unit area of the interface, the resistance is defined as

$$R_{t,c}'' = \frac{T_A - T_B}{q_x''}$$



The contact resistance may be minimized by

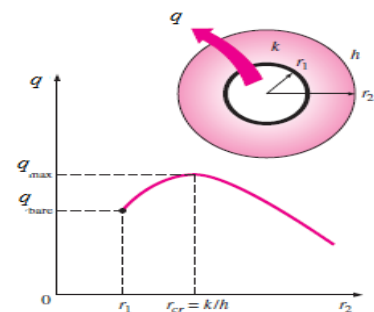
- *increasing the area of the contact spots.*
 - by increasing the joint pressure, and/or
 - by reducing the roughness of the mating surfaces.
- *selecting an interfacial fluid of large thermal conductivity.*

What is Critical radius of insulation? How is it defined for a cylindrical/spherical layer?

Adding insulation to a cylindrical pipe or a spherical shell increases the conduction resistance of the insulation layer but decreases the convection resistance of the surface because of the increase in the outer surface area for convection. For relatively smaller values of insulation thickness, the reduction in convection resistance outweighs the increase in conduction resistance. For larger values of insulation thickness the increase in conduction resistance outweighs the decrease in convection resistance. The outer radius of the insulation at which the sum of conduction and convection resistances is minimum is called the critical radius of insulation. The heat transfer through the insulation will be maximum at the critical radius of insulation.

$$r_{cr,cylinder} = \frac{k}{h}$$

$$r_{cr,sphere} = \frac{2k}{h}$$



Derivation

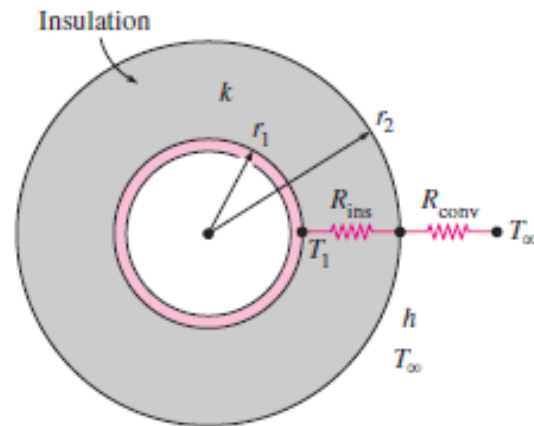
Consider a cylindrical pipe of outer radius r_1 whose outer surface temperature T_1 is maintained constant. The pipe is now insulated with a material whose thermal conductivity is k and outer radius is r_2 . The rate of heat transfer from the insulated pipe to the surrounding air (h, T_∞) can be expressed as

$$q = \frac{T_1 - T_\infty}{R_{ins} + R_{conv}} = \frac{T_1 - T_\infty}{\frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi L k} + \frac{1}{h(2\pi r_2 L)}}$$

The value of r_2 at which q reaches a maximum is determined from the requirement that, $\frac{dq}{dr_2} = 0$

Performing the differentiation and solving for r_2 yields the critical radius of insulation for a cylindrical body to be

$$r_{cr, cylinder} = \frac{k}{h}$$



For a spherical container of outer radius r_1 and temperature T_1 with insulation of material having thermal conductivity k and radius r_2 , the rate of heat transfer to the surrounding air can be expressed as,

$$q = \frac{T_1 - T_\infty}{R_{ins} + R_{conv}} = \frac{T_1 - T_\infty}{\frac{r_2 - r_1}{4\pi r_1 r_2 k} + \frac{1}{h(2\pi r_2 L)}}$$

The critical radius of insulation for the spherical container can be obtained as in the previous case for a cylindrical pipe.

$$r_{cr, sphere} = \frac{2k}{h}$$

What is a lumped system?

In heat transfer analysis, some bodies are observed to behave like a “lump” whose interior temperature remains essentially uniform at all times during a heat transfer process. The temperature of such bodies can be taken to be a function of time only, $T(t)$. Such bodies are called lumped systems.

The lumped system assumption is applicable when the Biot number (the ratio of conduction resistance within the body to convection resistance at the surface of the body) is less than or equal to 0.1.

$$Bi = \frac{hL_c}{k} < 0.1$$

Derivation

Consider a body of arbitrary shape of mass m , volume V , surface area A_s , density, ρ , and specific heat C_p initially at a uniform temperature T_i . At time $t=0$, the body is placed into a medium at temperature, T_∞ and heat transfer takes place between the body and its environment, with a heat transfer coefficient h . Let $T_i > T_\infty$. Assume the body to be a lumped system.

During a differential time interval dt , the temperature of the body decreases by a differential amount dT . An energy balance of the solid for the time interval dt can be expressed as

$$-mc_p dT = hA_s (T - T_\infty) dt$$

Noting that $m=\rho V$ and $dT = d(T - T_\infty)$ since $T_\infty = \text{constant}$, the above equation can be rearranged as

$$-\rho V c_p d(T - T_\infty) = hA_s (T - T_\infty) dt$$

$$\frac{d(T - T_\infty)}{(T - T_\infty)} = -\frac{hA_s}{\rho V c_p} dt$$

Integrating from $t=0$ to t , during which the temperature of the body changes from, T_i to T respectively.

$$\ln \frac{(T - T_\infty)}{(T_i - T_\infty)} = -\frac{hA_s}{\rho V c_p} t$$

Taking the exponential of both sides and rearranging,

$$\frac{(T - T_\infty)}{(T_i - T_\infty)} = e^{-\frac{hA_s}{\rho V c_p} t}$$